Revisiting the ATRAC3 Codec

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Abstract

This note documents `at3rs`, a Rust workbench implementation of Sony's ATRAC3 perceptual audio codec. ATRAC3 is a constrained-bitrate transform codec designed for the MiniDisc Long Play and Sony Memory Stick families of products, operating at 132/105/66 kbps on 44.1 kHz stereo input. We describe the codec's theoretical formulation—a four-band quadrature mirror filter (QMF) tree feeding per-band Modified Discrete Cosine Transforms (MDCT), followed by block-floating quantization with a perceptually-weighted bit allocation and Huffman-coded mantissas—and map every stage to its corresponding implementation in the current source tree. The aim is a self-contained reference: enough math to recover the algorithm from first principles, and enough code citations to locate the constants, tables, and decision rules used by the current encoder.

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1 Introduction

ATRAC3 (*Adaptive TRansform Acoustic Coding 3*) is a perceptual audio codec belonging to the same family as MPEG-1 Layer III, AAC, and Dolby AC-3: a critically-sampled, transform-based coder that quantizes spectral coefficients under a perceptual masking budget, then losslessly packs the resulting bitstream. It was deployed by Sony in MiniDisc-LP, ATRAC Audio Device (Walkman), and certain PlayStation Portable titles, and remains the format of choice for several reverse-engineered legacy toolchains.

`at3rs` is a from-scratch Rust implementation of the encoding side of ATRAC3. The current encoder emits RIFF/WAVE ATRAC3 files that are accepted by Sony's PSP ATRAC tool; exact quality parity with Sony's encoder is still an active target. The decoding path exists for matrix evaluation but is not yet usable quality. The workbench is structured around a small number of files:

```
src/atrac3.rs   - filterbank, MDCT, gain/tonal
     experiments, BFU plan, packer
src/atrac3/     - channel state, constants,
    config, debug helpers
src/codec.rs    - codec-agnostic I/O glue
src/dsp.rs      - shared helpers
src/eval.rs     - SNR/PSNR roundtrip metrics
eval/           - local matrix evaluation
    runner
src/gha.rs      - tonal-analysis support types
src/huffman.rs  - canonical Huffman tables
src/psychoacoustic.rs - shared PA helpers
```

The remainder of this document follows the encode path top to bottom: §2 sketches the frame layout and pipeline, §3 derives the QMF/MDCT analysis, §4 describes the block-floating spectral representation, §5 details the perceptual bit allocator, §6 covers entropy coding and bitstream packing, and §7 discusses unfinished components, public-surface decisions, and evaluation.

2 Pipeline Overview

ATRAC3 operates on fixed-size frames of N=1024 PCM samples per channel at F_s=44,100 Hz, corresponding to a frame duration of ≈ 23.2 ms. The frame size is *not* a free parameter: it is baked into the QMF delay line, the MDCT length, and the bitstream container format. The supported stream bitrates correspond to per-channel byte budgets

$$B_{\mathrm{ch}} \in \{96,\ 152,\ 192\}\ \text{bytes/frame}, \qquad (1)$$

i.e. 66, 105, and 132 kbps for mono, doubled for stereo. Selection logic lives in `choose_atrac3_block_align` at

`src/encoder.rs`, which simply rounds the requested bitrate to the nearest valid budget.

The encoder is a strict pipeline (`Atrac3Context::encode_frame`):

1. Convert interleaved `i16` PCM to planar $f32$ in $[-1, 1]$.
2. Per channel, run the two-level QMF analysis tree to produce four 256-sample subbands.
3. Per subband, run a length-512 MDCT (with 256-sample overlap) to obtain 256 spectral coefficients per band, for 1024 coefficients total per channel.
4. Optionally detect gain-control points and experimental tonal components. These paths are controlled by explicit `EncoderConfig` fields and CLI flags, and are disabled by default.
5. Group the residual 1024 coefficients into 32 Block Floating Units (BFUs); for each BFU, choose a quantizer selector $s \in \{0, \ldots, 7\}$ and a scale-factor index $k \in \{0, \ldots, 63\}$.
6. Iterate the selector assignment under a perceptually-weighted budget search, then run shaping/refinement passes until the total bit count fits the per-channel budget B_{ch}.
7. Pack the BFU header, scale-factor indices, and quantized mantissas into the per-channel sound unit, choosing between Huffman (VLC) and fixed-length (CLC) mantissa coding on a frame-global basis.

The command-line tool also exposes an experimental `-d` path so the evaluation matrix can exercise `decode_frame`. That decoder is known to be poor quality today and is diagnostic only; it is not a supported listening-quality decoder.

3 Filterbank: QMF Tree and MDCT

3.1 Two-stage QMF analysis

ATRAC3 uses a Polyphase Quadrature Filterbank (PQF) tree that splits the input into four contiguous, equal-bandwidth subbands. The prototype filter is a 48-tap symmetric FIR encoded as a 24-tap half filter at `src/atrac3.rs:16`, expanded in `QMF_WINDOW` (`src/atrac3.rs:25`) by mirroring and a factor-of-two gain compensation:

$$h[n] = h[47-n] = 2\,h_{\mathrm{half}}[n], \quad n = 0, \ldots, 23. \tag{2}$$

Let $x[n]$ be the input. A single QMF stage produces low- and high-pass half-rate outputs

$$y_\ell[m] = \sum_{i=0}^{23} \Big(h[2i]\,x[2m+47-2i] + h[2i+1]\,x[2m+46-2i] \Big), \tag{3}$$

$$y_h[m] = \sum_{i=0}^{23} \Big(h[2i]\,x[2m+47-2i] - h[2i+1]\,x[2m+46-2i] \Big), \tag{4}$$

i.e., a polyphase decomposition of the prototype h where the high-band is formed by alternating-sign modulation of the odd-indexed taps. The implementation lives in `pqf_forward` at `src/atrac3.rs:1633` and maintains a 46-sample history per stage in `Atrac3ChannelUnit::an_delay_buf{1,2,3}`.

The full analysis applies three stages organised as a tree (`analyze_frame`, `src/atrac3.rs:480`):

1. Stage 1: $1024 \to 512\,\mathrm{lo} \parallel 512\,\mathrm{hi}$.
2. Stage 2a: $512\,\mathrm{lo} \to 256\,\mathrm{B}_0 \parallel 256\,\mathrm{B}_1$.
3. Stage 2b: $512\,\mathrm{hi} \to 256\,\mathrm{B}_3 \parallel 256\,\mathrm{B}_2$.

Note that the second split of the high band yields the four output bands in the order $\mathrm{B}_0, \mathrm{B}_1, \mathrm{B}_2, \mathrm{B}_3$ once B_2 and B_3 are swapped: the inner high-band split's own high-pass output corresponds to band B_3, the highest frequency. This swap is applied unconditionally in `analyze_frame`. At $F_s = 44.1\,\mathrm{kHz}$ the four bands cover [0, 5.5125], [5.5125, 11.025], [11.025, 16.5375], and [16.5375, 22.05] kHz.

3.2 Per-band MDCT

Each 256-sample subband is then transformed by a length-$2N$, N=256 MDCT with 50% overlap. Recall that the analysis MDCT is

$$X[k] = \sum_{n=0}^{2N-1} w_a[n]\,x[n]\cos\Big(\tfrac{\pi}{N}(n+\tfrac{N+1}{2})(k+\tfrac{1}{2})\Big), \tag{5}$$

for $k = 0, \ldots, N-1$, with the standard time-domain aliasing cancellation (TDAC) constraint $w_a[n]\, w_s[n] + w_a[n+N]\, w_s[n+N] = 1$ for the overlap-add to recover the original samples after inverse MDCT.

Window choice. ATRAC3 uses an asymmetric design where the analysis window w_a is the lifted half-sine

$$w_a[n] = 1 + \sin\Big(\big(\tfrac{n+0.5}{256} - 0.5\big)\,\pi\Big), \quad n = 0, \ldots, 255, \tag{6}$$

encoded directly in `ENCODE_WINDOW` (`src/atrac3.rs:54`). The synthesis window is derived from w_a via the canonical Princen–Bradley dual

$$w_s[n] = \frac{2\, w_a[n]}{w_a[n]^2 + w_a[255-n]^2}, \tag{7}$$

which is precomputed as `DECODE_WINDOW` (`src/atrac3.rs:62`) and ensures TDAC perfect reconstruction. The auxiliary half-sine `MDCT_WINDOW` (`src/atrac3.rs:44`) is used only by the FFT-based MDCT kernel below.

Fast MDCT via FFT-128. The implementation factors the length-512 MDCT through a length-$N/4 = 128$ complex FFT (provided by `rustfft`). The standard derivation is to fold $2N$ reals into $N/2$ complex samples using the symmetry of the cosine basis. `at3rs` computes pre- and post-twiddle tables

$$\alpha = \frac{2\pi}{8 \cdot 2N}, \quad \omega = \frac{2\pi}{2N}, \tag{8}$$

$$c_i = s_{\text{scale}} \cos(\omega i + \alpha), \tag{9}$$

$$s_i = s_{\text{scale}} \sin(\omega i + \alpha), \tag{10}$$

with $s_{\text{scale}} = \sqrt{1/(2N)}$ for the forward direction (`src/atrac3.rs:344`) and $s_{\text{scale}} = \sqrt{N/(2N)} \cdot \sqrt{2}$ effectively for the inverse. The pre-twiddle folds the input as

$$r_0 = x[3N/2 - 1 - i] + x[3N/2 + i], \tag{11}$$

$$i_0 = x[N/2 + i] - x[N/2 - 1 - i], \tag{12}$$

$$Z[i/2] = (r_0 c_i + i_0 s_i,\ i_0 c_i - r_0 s_i), \tag{13}$$

for $i \in [0, N/2)$ (`src/atrac3.rs:358`), with the symmetric re-folding for $i \in [N/2, N)$. After the FFT the post-twiddle stores $X[2j] = -r_j c_j - i_j s_j$ and $X[N-1-2j] = -r_j s_j + i_j c_j$ (`src/atrac3.rs:376`).

Odd-band reversal. Bands B_1 and B_3 are spectrally reversed prior to or after their MDCT (`odd_band == true`, `src/atrac3.rs:389`). This compensates for the alternating sign introduced by each polyphase QMF stage so that the concatenated $4 \times 256 = 1024$-bin spectrum has a monotonic frequency mapping when reassembled.

4 Spectral Grouping and Block-Floating Quantization

4.1 Block Floating Units

The 1024 MDCT coefficients are partitioned into 32 non-uniform *Block Floating Units* (BFUs) defined by the boundary table `ATRAC3_SUBBAND_TAB` at `src/atrac3.rs:99`:

$$[0, 8, 16, 24, 32, 40, 48, 56,\ 64, 80, 96, 112, 128, 144, 160, 176,$$

$$192, 224, 256, 288, 320, 352, 384, 416,\ 448, 480, 512, 576, 640, 704, 768, 896,$$

The widths grow approximately geometrically with frequency, mimicking the critical-band structure of human hearing: 8-bin BFUs in the lowest $\sim$350 Hz, 128-bin BFUs in the highest $\sim$5 kHz.

Within each BFU b of width L_b, the spectral block $\mathbf{X}_b = (X[i])_{i \in [\text{start}_b, \text{end}_b)}$ is encoded as a single shared scale factor σ_b (the *block float exponent*) and L_b signed mantissas $\mathbf{q}_b$:

$$\widehat{X}[i] = q[i]\, \frac{\sigma_b}{Q_s}, \tag{14}$$

where Q_s is the per-selector quantizer maximum (Table 1).

4.2 Scale-factor table

The scale-factor codebook contains 64 values logarithmically spaced $\frac{1}{3}$ bit apart:

$$\sigma[k] = 2^{k/3-21}, \qquad k = 0, \ldots, 63, \tag{15}$$

implemented in the ATRAC3 constant table. The minimum representable level is $2^{-21} \approx 4.8 \cdot 10^{-7}$ and the maximum is $2^{42/3} \approx 100.3$. Each BFU uses a 6-bit field to transmit k.

4.3 Quantizer selectors

A BFU's bit budget is determined by its *selector* $s_b \in \{0, \ldots, 7\}$, where $s_b = 0$ means the entire BFU is treated as zero (no mantissas

s	Q_s	$\lvert q\rvert_{\max}$	nominal bits/coeff
1	1.5	1	∼2 (paired)
2	2.5	2	3
3	3.5	3	3
4	4.5	4	4
5	7.5	7	4
6	15.5	15	5
7	31.5	31	6

Table 1: Per-selector quantizer parameters used by the BFU planner.

transmitted, no scale factor transmitted). The non-zero selectors define quantizer maxima

For a BFU with chosen σ_b and s_b, each coefficient is encoded as

$$q[i] = \mathrm{clip}\left(\mathrm{round}\left(\frac{X[i]}{\sigma_b}\, Q_{s_b}\right),\ -|q|_{\max},\ |q|_{\max}\right). \tag{16}$$

4.4 Energy-preserving rounding

A naive midpoint round to nearest minimises per-coefficient L^2 error but biases the per-block reconstructed energy: dense small-magnitude coefficients tend to round to 0, reducing total spectral power and producing audibly hollow textures. `quantize_block_energy_adjusted` addresses this by collecting *borderline* coefficients, those whose unrounded value lies near a half-integer, and iteratively flipping their rounding direction toward whichever choice (up or down by one quantum) brings reconstructed block energy closer to $\sum_i X[i]^2$. The candidate list is sorted by how close each value is to the half-integer boundary so that the cheapest swaps are applied first. The result is a quantizer that is no longer strictly minimum-MSE but preserves block-level loudness. This energy-adjusted rounding is the default; `AT3RS_DISABLE_SF_SEARCH` and related configuration flags can disable some of the surrounding search machinery for experiments.

5 Perceptually-Weighted Bit Allocation

The encoder must choose $(s_b, k_b)_{b=0}^{31}$ for each channel under the constraint that the total emitted bits do not exceed $8\,B_{\mathrm{ch}}$. At the high end of the allowed bitrate (132 kbps) the allocator can afford a near-uniform selector pattern; at 66 kbps it is forced into aggressive masking decisions.

5.1 Loudness and absolute threshold of hearing

The first ingredient is the *equal-loudness curve* $\mathcal{L}[i]$, a 1024-bin weighting precomputed at `src/atrac3.rs:72` as

$$f_i = \frac{(i+3)\cdot F_s/2}{1024}, \tag{17}$$

$$t_i = -10(\log_{10} f_i - 3.5)^2 + 3 - \frac{f_i}{3000}, \tag{18}$$

$$\mathcal{L}[i] = 10^{0.1\,t_i}. \tag{19}$$

This is a smooth quadratic-in-log frequency bowl peaked near 3 kHz with a high-frequency rolloff, used to weight per-bin energy when computing the channel loudness norm Λ_{ch} below.

The second ingredient is the per-BFU *absolute threshold of hearing* (ATH), $T_{\mathrm{ATH}}[b]$. The Frank approximation `ath_formula_frank` (`src/atrac3.rs:124`) provides per-bin SPL thresholds via a 140-entry tabulated curve indexed in $\log_{10}(f/100)$ with frac-linear interpolation:

$$\ell = 40\log_{10}(f/100), \tag{20}$$

$$\mathrm{ATH}_{\mathrm{dB}}(f) = \mathrm{TAB}[\lfloor\ell\rfloor]\,(1-\{\ell\}) + \mathrm{TAB}[\lfloor\ell\rfloor+1]\{\ell\}. \tag{21}$$

After subtracting the 100 dB calibration offset and the $0.015\,(f/1\,\mathrm{kHz})^2$ high-frequency tilt (`calc_ath_spec`, `src/atrac3.rs:147`), the minimum threshold inside each BFU is converted back to linear amplitude and stored in `ATRAC3_ATH_BFU` (`src/atrac3.rs:83`).

The current implementation does *not* compute simultaneous (in-band) or non-simultaneous (temporal) masking thresholds. The masking model is the absolute threshold scaled by a tracked global loudness, described next.

5.2 Tracked loudness norm

Per-frame loudness in channel c is

$$L_c = \sum_{i=0}^{1023} X_c[i]^2\,\mathcal{L}[i], \tag{22}$$

(`frame_channel_loudness`, `src/atrac3.rs:819`). A first-order exponential filter tracks the global loudness across frames:

$$\Lambda_t = 0.98\,\Lambda_{t-1} + 0.01 \sum_c L_{c,t}, \qquad (23)$$

and the masking threshold for BFU b becomes

$$T_b = T_{\mathrm{ATH}}[b] \cdot \frac{\Lambda_t}{\Lambda_{\mathrm{factor}}}, \quad \Lambda_{\mathrm{factor}} = 0.006. \quad (24)$$

A BFU whose energy falls below T_b is forced to selector 0 and the mantissas/scale factor are not coded. The constants 0.98, 0.01, and $\Lambda_{\mathrm{factor}}$ (`src/atrac3.rs:122`) were tuned empirically on the payload fixtures in `fixtures/`.

5.3 Per-BFU importance score

Each BFU is assigned a scalar importance combining bulk energy with peak prominence:

$$I_b = \left(\sqrt{E_b} + 8\,M_b\right) \cdot \left(1 + 0.2\,\phi_b\right) \cdot \left(1 + 0.05\,\beta_b\right), \qquad (25)$$

where $E_b = \sum_{i \in b} X[i]^2$, $M_b = \max_i |X[i]|$, ϕ_b is the entry of `ATRAC3_FIXED_ALLOC_TABLE` for BFU b (`src/atrac3.rs:107`), and $\beta_b \in \{0, 1, 2, 3\}$ is the QMF band index. The peak weighting $8M_b$ biases the allocator toward preserving tonal/transient peaks even when bulk energy is small.

5.4 Initial selector heuristic

A scale-factor variance term ρ_b measures how broadband-flat the block is:

$$\rho_b = \min\left(\sqrt{\mathrm{Var}(k_b)},\ 14\right) / 14, \qquad (26)$$

ranging from 0 (all coefficients near the same magnitude) to 1 (highly heterogeneous). The initial selector is then computed as

$$s_b = \mathrm{clip}\left(\left\lfloor \rho_b \tfrac{k_b}{D_b} + (1-\rho_b)\,\phi_b - \delta + \gamma_b \right\rfloor,\ 0,\ 7\right), \qquad (27)$$

where D_b is a block-dependent divisor (`selector_spread_divisor`, `src/atrac3.rs:686`) ranging from 2.6 at low frequency to 6.0 at high frequency, δ is a global shift parameter swept during the budget search, and γ_b bundles three additional perceptual biases:

- $\gamma_b^{(I)}$, an *importance boost* from Eq. 25 stepping over the thresholds $\{3.5, 6.0, 10.0\}$ to $\{0, 0.5, 1.0, 2.0\}$ (`importance_boost`, `src/atrac3.rs:665`),
- $\gamma_b^{(M)}$, an *average magnitude bias* based on $\log_{10} \overline{|X|_b}$ (`avg_mag_bias`, `src/atrac3.rs:649`), giving negative γ for very quiet blocks and positive for very loud ones,
- $\gamma_b^{(S)}$, a *shape hint* (`ATRAC3_SELECTOR_SHAPE_HINT`, `src/atrac3.rs:113`) decaying from 7 in the lowest BFUs to 0 at the top, encoding the prior that high frequencies generally tolerate coarser quantization.

The clip is special-cased so a non-zero s_b is never mapped to 0 implicitly — selector 0 only happens through the explicit masking check of Eq. 24.

5.5 Budget search and refinement

The current planner performs a binary search over a global selector shift $\delta \in [-8, 20]$, recomputing the full 32-BFU plan for each trial and retaining the best plan whose encoded bit count does not exceed the per-channel budget. Before the search, the effective budget is reduced by header overhead for gain/tonal side data and, when VLC is enabled, by a small configurable safety margin (`EncoderConfig::vlc_bit_safety`, default 32 bits). This keeps later coding-mode decisions from accidentally overflowing the sound-unit payload.

Once the binary search chooses a feasible plan, the encoder runs a sequence of shaping passes. The default pass, `fit_bfu_plan_to_budget`, tries local selector changes and accepts changes that improve weighted reconstruction error while remaining under budget. Additional experimental passes can be enabled from `EncoderConfig`:

- `enable_tail_prune`: enforce a non-increasing high-frequency selector tail.
- `experimental_concentrate_bits`: prune low-energy high-frequency BFUs when they only receive very low precision.
- `experimental_tonal_prune`: preserve a small set of dominant BFUs on concentrated tonal frames and prune weaker neighbours.
- `enable_tail_sf_smooth`: smooth tail scale-factor indices to avoid abrupt high-frequency gain jumps.

- `disable_reference_shape`: opt out of the default pass that nudges selectors toward the reference-shaped high-frequency allocation prior.
- `experimental_rebalance_bfus`: try donor/receiver precision swaps and keep a candidate if weighted error improves.
- `cap_high_bfus`: clear BFUs above a fixed high-frequency cutoff for debugging.

After every shaping pass, the planner recomputes the channel coding mode and re-runs budget fitting. This makes the allocation stage a small rate–distortion search rather than a single feed-forward selector formula.

6 Entropy Coding and Bitstream

6.1 Coding mode

Mantissas are emitted in one of two modes, signaled by a single bit:

- VLC (`coding_mode = 0`): canonical Huffman codes, selector-dependent, with selector 1 using a paired-symbol alphabet over (q_a, q_b) so that highly-quantized blocks (where most coefficients are $0, \pm 1$) get sub-bit-per-coefficient rates.
- CLC (`coding_mode = 1`): fixed-length two's-complement codes at the nominal width of Table 1.

The selection is made per channel by `channel_prefers_vlc`. VLC is disabled when `EncoderConfig::force_clc` or `EncoderConfig::disable_vlc` is set. Otherwise the encoder optionally rejects plans whose non-zero selectors exceed `EncoderConfig::vlc_max_selector`; for the remaining plans it chooses VLC whenever the summed spectral VLC bit count is lower than the summed CLC bit count. The global budget search already subtracts the configurable VLC safety margin described above, so the mode decision itself is a direct CLC-versus-VLC bit-count comparison.

6.2 Huffman tables

The seven Huffman codebooks (one per non-zero selector) are stored as (symbol, length) pairs in `ATRAC3_HUFFTABS_RAW` (`src/huffman.rs:131`). The encoder materialises them into a canonical form (`src/huffman.rs:77`) that assigns codes by ascending (length, symbol) order, matching the FFmpeg reference. Symbols for selectors 2–7 are derived from a signed magnitude via

$$\mathrm{sym}(q) = \begin{cases} 2|q| & q \geq 0 \\ 2|q| + 1 & q < 0 \end{cases} - 1, \qquad (28)$$

(`scalar_vlc_symbol`, `src/atrac3.rs:1414`). Selector 1 maps the 9 possible $(q_a, q_b) \in \{-1, 0, +1\}^2$ pairs to a 0–8 alphabet directly (`selector1_vlc_symbol`, `src/atrac3.rs:1308`).

6.3 Per-channel sound unit layout

Each channel's sound unit is laid out as (`write_channel_sound_unit`, `src/atrac3.rs:1336`):

```
[ 6 b ] sync prefix      = 0x28
[ 2 b ] frame_type       = 3
[12 b ] reserved
[ 5 b ] joint_stereo_info= 0
[ 5 b ] num_blocks - 1
[ 1 b ] coding_mode      (0=VLC, 1=CLC)
[ 3 b ] selector[0..num_blocks]   each
[ 6 b ] sf_idx[0..num_blocks]     for nonzero
    sel.
[ var ] mantissas (VLC or CLC)
[ pad ] zero-fill to channel budget
```

Streams are wrapped in a RIFF/WAVE container with format tag 0x0270 (ATRAC3) and a `fact` chunk advertising the per-channel sample count and the 1024 samples-per-frame value (`write_at3_riff`, `src/main.rs:244`). An optional `smpl` chunk carries loop points for streams used in interactive applications.

7 Joint Stereo, Gain/Tonal Coding, and Evaluation

7.1 Joint stereo (not implemented)

The reference ATRAC3 bitstream supports a joint-stereo mode (intensity-coded high frequencies above a per-frame split point with a single transmitted scale per coupled BFU). `at3rs` reserves the `joint_stereo_info` field but always transmits 0, encoding each channel independently. This is consistent with the FFmpeg decoder behaviour for `joint_stereo_info == 0` but represents a hard ceiling on stereo bitrate efficiency: a true joint-stereo path could save $\sim 20\%$ of the high-frequency bit budget on correlated material.

7.2 Gain control and tonal coding

ATRAC3 sound units can carry gain-control side data and tonal components in addition to the residual MDCT spectrum. `at3rs` now contains experimental implementations for both, but they are deliberately disabled in the default encoder configuration because neither path is yet a quality win on the fixture corpus.

Gain detection is enabled with `EncoderConfig::experimental_gain` or the newer `EncoderConfig::experimental_gain_v2`; the CLI exposes the v2 path as `-enable-gain-v2`. The encoder searches each QMF band for short-time energy changes and writes gain points before MDCT packing. When gain points are present, the corresponding side-information bits are subtracted from the BFU budget and the analysis windows are modulated so that the residual spectrum matches the transmitted gain envelope.

Tonal components are enabled with `EncoderConfig::experimental_tonal_components`; the CLI exposes this as `-enable-tonal-components`. The detector looks for locally dominant spectral peaks, quantizes short runs with high precision (selector 7), groups compatible components, writes them in the tonal substream, and subtracts their reconstructed contribution from the residual before BFU planning. Separately, the default planner includes a *tonal-frame boost*: if energy is concentrated in a few BFUs, the planning spectrum is temporarily boosted and the written scale factors are offset back down. This is not the same as emitting tonal syntax; it is a bit-allocation heuristic for exposed tonal material.

7.3 Evaluation

`src/eval.rs` provides `evaluate_pair`, an offset-search quality scorer used by the Rust test harness. Given an original PCM signal x and a reconstructed signal $\hat{x}$, the function searches an offset window $[0, W]$ (default $W = 4096$) and reports

$$\mathrm{SNR}_{\mathrm{dB}}(\tau) = 10 \log_{10} \frac{\sum x[n]^2}{\sum (x[n] - \hat{x}[n+\tau])^2}, \tag{29}$$

$$\mathrm{PSNR}_{\mathrm{dB}}(\tau) = 20 \log_{10} \frac{32767}{\mathrm{RMSE}(\tau)}, \tag{30}$$

selecting the $\tau^\star$ maximising SNR. This compensates for the frame-aligned encoding/decoding delay introduced by the QMF and MDCT stages without a separate *a priori* delay estimate. `tests/quality.rs` and `tests/payload_quality.rs` use the metric to gate regressions on the fixture corpus in `fixtures/`.

For local listening and cross-implementation debugging, the primary workflow is now `eval/run_eval.py`. It evaluates an encoder/decoder comparison matrix for each fixture:

encoder	decoder
`at3rs`	`at3rs`
`at3rs`	Sony `psp_at3tool.exe`
Sony `psp_at3tool.exe`	`at3rs`
Sony `psp_at3tool.exe`	Sony `psp_at3tool.exe`

The script writes decoded WAVs, a CSV summary, and a self-contained HTML report with base64-inlined metric plots under `output/<git-ref>/eval/<input-dir>/`. It can also run ViSQOL and PEAQ when the third-party binaries are present. This matrix is intentionally diagnostic: `at3rs->sony` measures the current encoder through Sony's decoder, `sony->sony` is the local reference loop, and any cell ending in `at3rs` tracks the known-broken at3rs decoder.

Figures 1–4 show the most recent full-fixture matrix available at the time of writing, measured on 2026-04-30 from `output/main@6775023-dirty/eval/fixtures/summary.csv`. These plots are intended as a dated snapshot, not a permanent benchmark: the output directory is keyed by git ref so future runs can be compared against the exact source revision that produced them.

8 Implementation Notes and Limitations

Decoder. The `-d` path parses RIFF/WAVE ATRAC3 files with format tag 0x0270, extracts frame payloads using the container block alignment, and writes PCM WAV output. It is kept wired into the CLI and eval runner so decoder progress can be measured, but current decoded PCM is not usable quality: recent matrix runs show the at3rs-decoder cells near 0 dB SNR while Sony decoding of the same at3rs

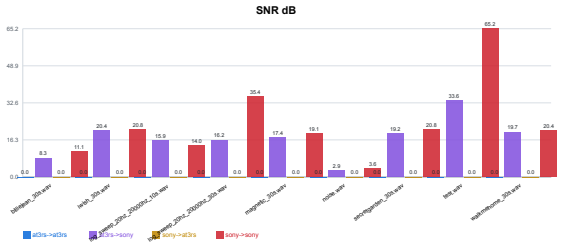

Figure 1: Current round-trip SNR by fixture and encoder/decoder cell. The `at3rs->sony` cell measures the at3rs encoder through Sony's decoder, while `sony->sony` is the local Sony-tool reference loop; cells decoded by `at3rs` expose decoder work still to be done.

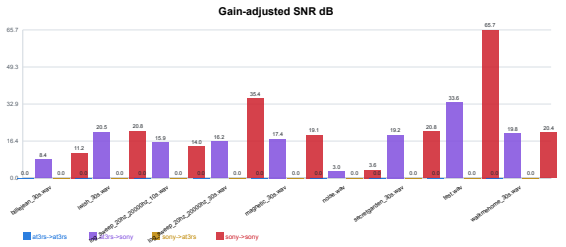

Figure 2: Current gain-adjusted SNR by fixture and encoder/decoder cell. The gain-adjusted score compensates for scalar output-level differences before computing residual energy.

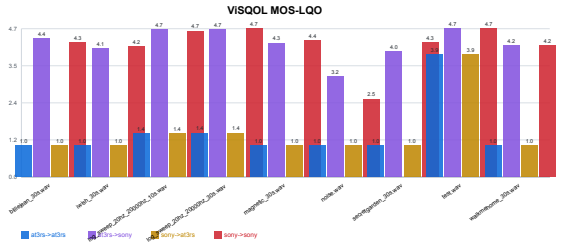

Figure 3: Current ViSQOL MOS-LQO by fixture and encoder/decoder cell. Higher scores are better; this metric is useful for trend tracking but does not fully capture crackle, ringing, or pitch-related artifacts.

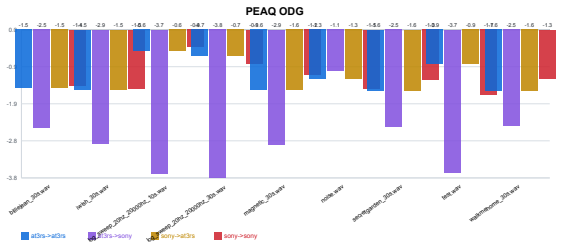

Figure 4: Current PEAQ ODG by fixture and encoder/decoder cell. ODG ranges from 0 for imperceptible impairment toward −4 for very annoying impairment; higher is better.

bitstream scores much higher. Decoder correctness is therefore a separate open problem, not evidence that the at3rs encoder bitstream is undecodable.

Sampling rates. Only 44.1 kHz input is supported. The QMF prototype filter and the loudness/ATH curves are tabulated for this rate; supporting 48 kHz or 32 kHz ATRAC3 variants would require regenerating these tables.

Tuning. The constants in Eq. 24–27 (0.98, 0.01, Λ_{factor}, the importance and shape-hint vectors, the divisor steps in `selector_spread_divisor`) are fitted to the payload fixtures and are not derived from a principled rate-distortion optimisation. The shape of the bit-allocation pipeline mirrors FFmpeg's encoder `atrac3enc.c` but the exact numeric tradeoffs differ.

ATRAC3+. `src/atrac3plus_data.rs` contains the lookup tables (`ATRAC3P_WL_CBS`, `ATRAC3P_CT_CBS`, `ATRAC3P_SPECTRA_CBS`, etc.) needed by the wider ATRAC3+ toolchain, but no ATRAC3+ encoding or decoding is implemented in this workbench. They are reserved for a future companion module.

Acknowledgements

This workbench builds on prior reverse-engineering work on ATRAC3 that is reflected in the FFmpeg source tree (`libavcodec/atrac3*.c`). The QMF prototype filter coefficients and the seven Huffman tables in this document are reproductions of those artefacts.

A Reference Constant Map

Symbol	Value / source	Location
F_s	44,100 Hz	required
N	1024 samples/frame	`atrac3.rs`
B_{ch}	96/152/192 B	`main.rs:153`
h_{half}	24-tap	`atrac3.rs:16`
$\sigma[k]$	$2^{k/3-21}$	`atrac3.rs:36`
w_a	Eq. 6	`atrac3.rs:54`
w_s	Eq. 7	`atrac3.rs:62`
$\mathcal{L}$	loudness curve	`atrac3.rs:72`
T_{ATH}	per-BFU ATH	`atrac3.rs:83`
$\Lambda_{\mathrm{factor}}$	0.006	`atrac3.rs:122`
ϕ_b	fixed alloc	`atrac3.rs:107`
$\gamma_b^{(S)}$	shape hint	`atrac3.rs:113`

Table 2: Map from symbols used in this document to their definitions in the source tree.